



KerisRDNet: Mask-aware augmentation and residual dilated networks for cultural heritage blade classification

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ABSTRACT

Fine-grained recognition of cultural artifacts remains challenging because of the scarcity of annotated data, subtle intra-class differences, and heterogeneous imaging conditions. This study addresses these issues through a domain-specific deep learning pipeline, demonstrated on Indonesian keris classification across three tasks: *pamor* (27 classes), *dhapur* (42), and *tangguh* (5). The pipeline integrates background homogenization, orientation normalization, and YOLOv8-based blade cropping with mask-aware augmentation restricted to the blade regions. For classification, we propose KerisRDNet, which extends InceptionResNetV2 with Inception-Residual-Dilated (IRD) blocks and squeeze-and-excitation to model the elongated geometries and subtle forging motifs. Experiments show that baseline networks collapse under fine-grained settings, with macro-F1 near zero, whereas the proposed approach achieves 0.268 (*pamor*), 0.276 (*dhapur*), and 0.635 (*tangguh*) with Top-3 accuracy above 0.5 and AUC up to 0.853. Across three stratified resamplings, paired non-parametric tests (Wilcoxon signed-rank) indicated directionally consistent improvements; given the small number of repetitions ($n = 3$), these results are interpreted conservatively. These results demonstrate the feasibility of practically viable keris recognition as a decision-support tool for cultural heritage curation, while also offering a transferable workflow for low-data fine-grained recognition tasks.

1. Introduction

The keris, a traditional Indonesian dagger, occupies a distinctive position in Southeast Asian cultural heritage. Recognized by UNESCO as an Intangible Cultural Heritage since 2008 (UNESCO, 2008), it embodies artisanal craftsmanship and symbolic and ritual significance that has persisted for centuries. Key diagnostic attributes include the *pamor* — a layered forging pattern visible on the blade's surface, the *dhapur* — referring to blade morphology and stylistic form, and the *tangguh* — a categorization of historical provenance. These features provide curators and scholars with critical information for provenance studies, conservation, and cataloguing, while also contributing to the intangible heritage associated with ritual practices and identity (Salvemini et al., 2020; Wang & Lu, 2024).

Despite its cultural importance, keris classification remains dependent on manual expert appraisal. Specialists visually identify *pamor* motifs, *dhapur* forms, and *tangguh* eras based on subtle cues that require years of tacit expertise. These manual procedures are time-consuming, subjective, and often inconsistent across practitioners, especially in the absence of universally standardized guidelines (Hou et al., 2018; Hu et al., 2024). Moreover, declining familiarity with traditional artifacts

among younger generations amplifies the urgency of developing digital preservation tools that can assist experts and democratize access to keris knowledge (Konstantinov et al., 2012).

Computational efforts have begun to demonstrate the feasibility of automating the analysis of keris. Caesarendra et al. (2021) employed 3D scanning for geometric reconstruction, Wibawa et al. (2024) leveraged convolutional neural networks (CNNs) for *pamor* and *tangguh* classification, Sebatubun and Haryawan (2024) addressed *dhapur* recognition using a small curated dataset and CNN-based modeling, and Santoso and Hadjon (2015) applied edge-based segmentation with similarity metrics for motif matching. However, these studies face common constraints: limited dataset size, heterogeneous imaging conditions, and the absence of standardized preprocessing pipelines, which together hinder reproducibility and cross-collection scalability.

Comparable challenges are observed in other cultural heritage domains, such as edged weapons, ceramics, and textiles. In these cases, both fine-grained texture patterns and global shape features must be captured simultaneously for accurate recognition (Bu et al., 2025; Król et al., 2025; Malchiodi et al., 2025). Preprocessing through segmentation and background homogenization plays a critical role in suppressing

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spurious cues (Singh et al., 2025; Wang et al., 2023), whereas orientation correction has been shown to improve robustness in object recognition tasks (Lumini et al., 2021). Data scarcity remains another persistent obstacle, and recent studies have highlighted the benefits of domain-specific augmentation strategies that better simulate real-world acquisition conditions (Carvalho Ottoni & Cordeiro Ottoni, 2025; Sousa et al., 2025). At the architectural level, multiscale convolutions, dilated kernels, and channel recalibration modules have emerged as effective tools for the fine-grained recognition of complex surface structures (Liu et al., 2022). Recent advances in transformers (Dosovitskiy et al., 2021; Han et al., 2023) and interpretable attention mechanisms (Chefer et al., 2021) have further expanded the state of the art, although their potential in cultural heritage remains underexplored.

Similar bottlenecks are well documented in other fine-grained domains, such as bird species recognition (Wah et al., 2011; Zhang et al., 2014) and medical histopathology (Campanella et al., 2019; Coudray et al., 2018), where high intra-class similarity, low inter-class variability, and limited labeled datasets often cause standard CNN backbones to collapse. These parallels highlight that keris recognition is not an isolated problem but rather an instance of a broader class of fine-grained classification challenges under low-data, high-variability conditions.

Building on these insights, this study introduces an integrated deep learning pipeline tailored for keris classification. The pipeline incorporates domain-specific preprocessing steps, including U-Net-based background homogenization, principal component analysis (PCA) for orientation normalization with handle-down correction, and mask-aware augmentation restricted to blade regions. For classification, we propose KerisRDNet, an architecture explicitly designed for keris images. KerisRDNet extends InceptionResNetV2 with Inception-Residual-Dilated (IRD) blocks that employ multiscale separable convolutions, residual scaling, and a squeeze-and-excitation module. Unlike prior MRDI-style variants, KerisRDNet is optimized for the elongated geometries and fine-grained *pamor* textures characteristic of keris blades, aligning the architectural design with the specific diagnostic challenges of cultural heritage artifacts.

This study makes four contributions.

- We design a preprocessing workflow that standardizes object imagery (background, orientation, ROI cropping, and scale) to reduce nuisance variability.
- We introduce mask-aware augmentation that balances classes while restricting perturbations to diagnostically relevant regions.
- We propose KerisRDNet, a domain-aligned architecture that combines multiscale dilated convolutions, residual scaling, and channel recalibration for fine-grained recognition.
- We evaluate the full pipeline across three classification granularities via controlled ablations, modern baselines, and statistical tests, with keris recognition as a case study.

The remainder of this paper is organized as follows: Section 2 details the dataset, preprocessing, augmentation, and model architectures. Section 3 presents the experimental results and statistical validation. Section 4 interprets the contributions of each pipeline component, assesses the impact of KerisRDNet, and explores the implications for digital heritage. Section 5 outlines the study limitations and directions for future work. Finally, Section 6 concludes the paper.

2. Methods

2.1. Data acquisition and annotation

A curated dataset of keris images was compiled with expert-verified annotations across three single-label classification tasks: *pamor* (27 classes), *dhapur* (42 classes), and *tangguh* (5 classes). Examples include *pamor* motifs such as *Beras Wutah* and *Udan Mas*, *dhapur* forms such as *Brojol* and *Tilam Uphih*, and *tangguh* eras such as *Mataram* and *Majapahit*

Table 1

Summary of dataset composition across the three classification tasks.

Task	No. of classes	Train	Augmented	Validation	Test
<i>Pamor</i>	27	574	3693	86	86
<i>Dhapur</i>	42	420	2898	99	99
<i>Tangguh</i>	5	267	801	33	34

(see Appendix A, Table A.1–Table A.3, for complete class distributions). The corpus consisted of 574 *pamor*, 420 *dhapur*, and 267 *tangguh* images, which were later expanded through augmentation to over 4000 samples per task. Each physical keris artifact contributed between two and five photographs under varied poses and lighting, reflecting the heterogeneity typical of museum and private collections.

Annotations were performed by trained keris experts following established curatorial guidelines. The labels were cross-checked to ensure consistency across categories. In cases where initial disagreements arose among experts, labels were resolved through discussion and cross-verification; samples for which consensus could not be reached were excluded from the final dataset to avoid introducing ambiguous ground truth. To prevent data leakage, all images of the same artifact were grouped before dataset splitting. All data were collected with explicit curatorial consent and are treated as culturally governed materials rather than open-access benchmarks. Data handling followed ethical protocols for cultural heritage digitization, with custodians retaining authority over ownership and future use.

Table 1 summarizes dataset composition across the three tasks, highlighting pronounced class imbalance that motivates the subsequent augmentation strategy. These curated and annotated images constituted the raw input for the preprocessing stage. The overall computational pipeline, which covers data preparation, augmentation, modeling, and evaluation, is summarized in Fig. 1.

2.2. Preprocessing

Domain-specific preprocessing was applied to standardize the image presentation and suppress irrelevant variability, as summarized in Fig. 1. Each image was first managed as an RGB input with a neutralized EXIF orientation. Background removal was then performed using a U-Net model implemented in *rembg*, which generated alpha mattes that were subsequently refined by 3×3 morphological open/close operations. The resulting masks were composited onto a uniform white canvas (three-channel PNG, avoiding alpha), and an initial bounding box crop with a small margin was applied to remove empty regions.

To further reduce intra-class variance, vertical orientation was standardized via PCA on the object mask. Specifically, PCA was applied to the binary foreground mask obtained after background homogenization by treating all foreground pixel coordinates as a 2D point set and computing their covariance matrix. The first principal component (i.e., the eigenvector associated with the largest eigenvalue) was used as the dominant shape axis, and the image was rotated so that this axis aligned with the vertical direction. Because PCA is ambiguous up to 180° , we applied a rule-based handle-down correction by comparing the foreground pixel mass in the upper vs. lower half after rotation and flipping vertically when the denser region appeared in the upper half. After re-cropping to exclude residual background, images were resized proportionally to a height of 512 pixels and adjusted with horizontal padding or cropping to obtain a 512×512 square while maintaining a uniform white background without vertical padding.

Emphasis on diagnostically relevant regions was achieved through blade-focused cropping of the images. We trained a YOLOv8 detector on 980 annotated blade bounding boxes (880 training and 100 validation) with no artifact overlap across the splits. The detector used the YOLOv8x variant for 300 epochs with a batch size of 32, AdamW optimizer (initial learning rate 5×10^{-4} with cosine decay to $0.01\times$), dropout 0.10, mixed precision, and seed 42.

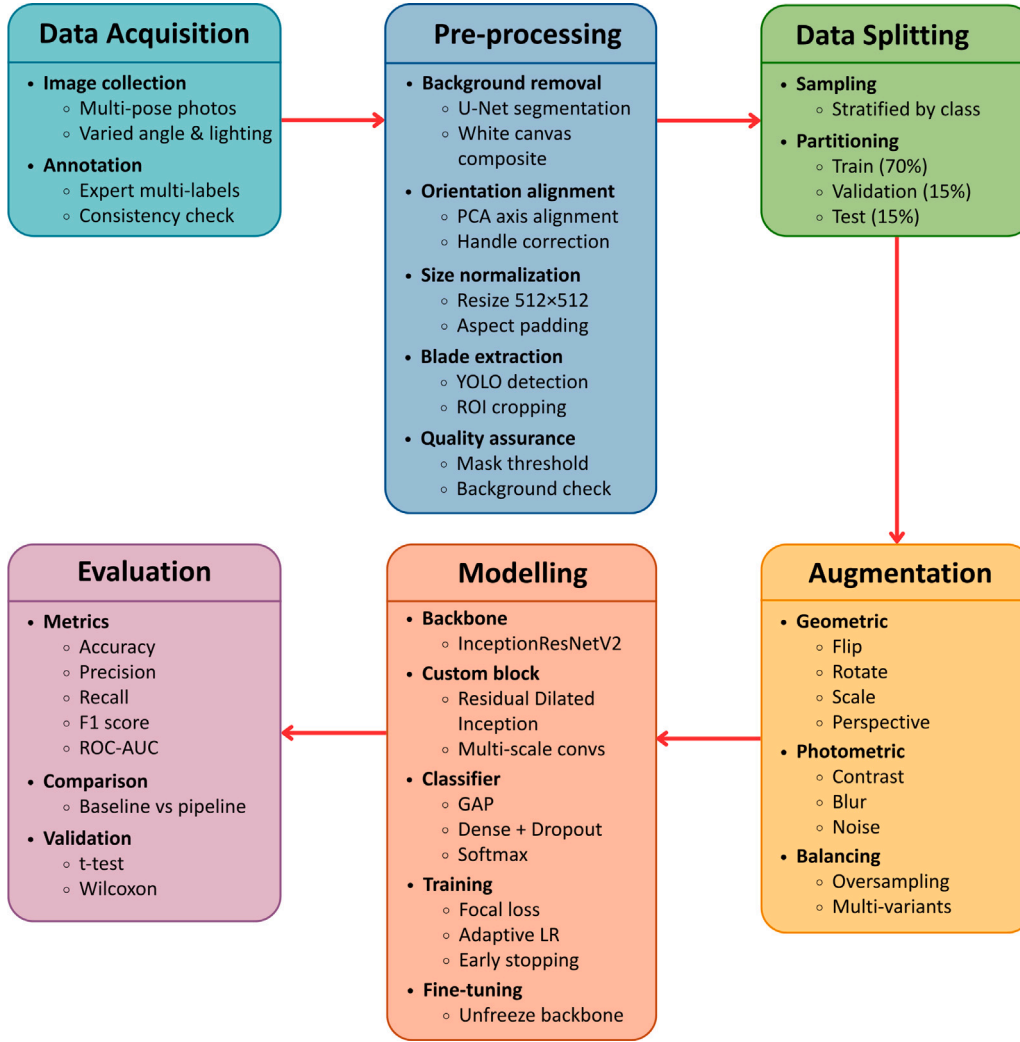


Fig. 1. Proposed pipeline for keris classification, including data acquisition, preprocessing, dataset splitting, augmentation, and final classification with KerisRDNet.

The best checkpoint achieved precision 0.998, recall 1.000, mAP_{50} 0.994, and mAP_{50-95} 0.961 on the validation set (Table 2). During inference, we applied non-maximum suppression ($IoU = 0.50$) and retained only the top-1 detection per image ($max_det = 1$), thereby removing any overlapping proposals. A symmetric 3% margin was added before cropping to avoid truncating the blade tip or guard, and the crop was resized to 512×512 . Detections with low confidence, abnormal aspect ratios (outside $[0.10, 10.0]$), or blade areas $< 1\%$ of the canvas were filtered out. Together with a final QA step that excluded failed segmentations and undersized masks, these procedures homogenized the visual domain, enforced consistent orientation and background, and concentrated learning on diagnostically salient blade regions and their *pamor* textures. After this stage, the dataset was partitioned into training, validation, and testing sets using a stratified artifact-level split to prevent leakage, as detailed in Section 2.7.

2.3. Dataset splitting

After preprocessing, the dataset was divided into training, validation, and testing partitions. The splits were stratified by class while grouping all images of the same physical artifact to prevent leakage across the sets. We adopted a 70%/15%/15% ratio, consistent across the three classification tasks (*pamor*, *dhapur*, and *tangguh*), as summarized in Table 1. This stratification ensured that minority classes

Table 2
YOLOv8 blade detector training hyperparameters.

Parameter	Value	Description
Variant	YOLOv8x	High capacity for thin edges and curved blades.
Epochs	300	Sufficient cycles to learn complex contours.
Image size	512	Preserves detail while fitting batch size 32.
Batch size	32	Balances stability and memory efficiency.
Optimizer	AdamW	Decoupled weight decay improved generalization.
Initial LR	5×10^{-4}	Conservative start to avoid divergence.
Final LR	0.01	Cosine decay to 1% of the initial rate.
Dropout	0.10	Light regularization against overfitting.
Patience	30	Early stop if validation stalls.
Seed	42	Ensures reproducibility across the runs.
AMP	True	Mixed precision for speed and lower VRAM.
Workers	4	Parallel data loading for a stable throughput.

were proportionally represented while preserving artifact-level independence between sets. Further details of the splitting protocol and its role in ensuring leakage-free evaluation are provided in Section 2.7.

2.4. Data augmentation

To mitigate class imbalance and enhance generalization, we designed a mask-aware augmentation pipeline that preserved a uniform white background and restricted photometric perturbations to the blade

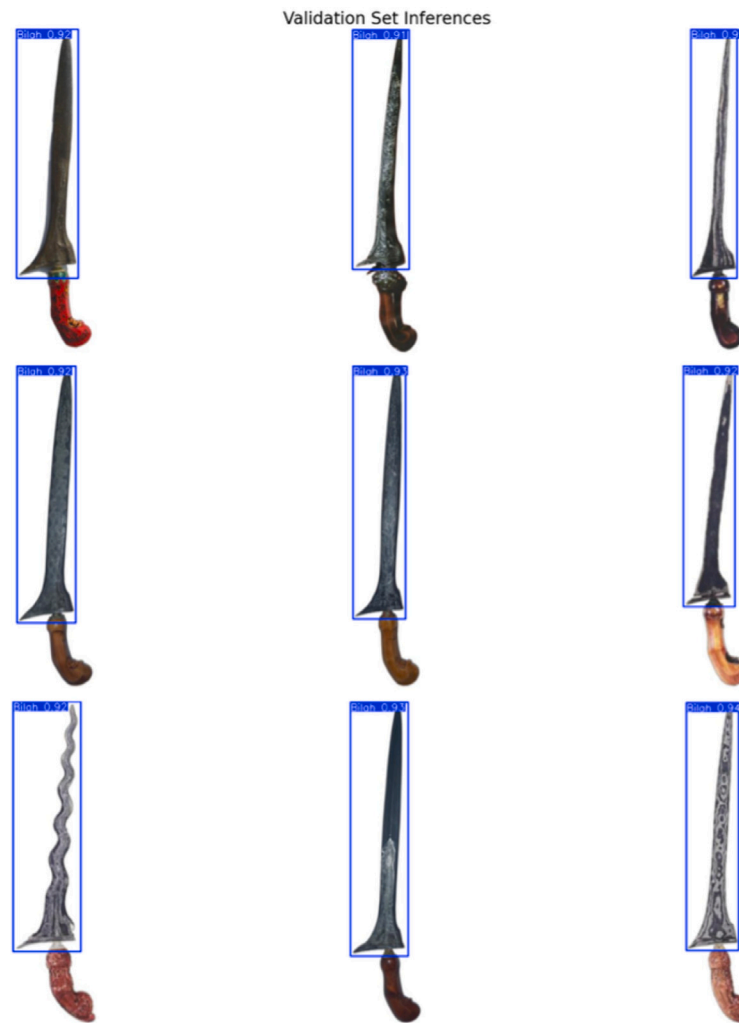


Fig. 2. Representative YOLOv8x detections of keris blades from the validation set, used for ROI extraction.

regions only. This strategy ensured that no spurious cues from non-blade areas were exploited during learning.

Augmentation began with pre-masking: binary blade masks were derived via near-white thresholding and were refined morphologically. The geometric transformations included horizontal flips ($p = 0.5$), small shift-scale-rotate operations ($p \approx 0.7$, within $\pm 6\%$ shift/scale and $\pm 5^\circ$ rotation), and light perspective distortions ($p \approx 0.2$). Photometric variants applied exclusively to blade pixels comprised brightness/contrast adjustments ($\pm 20\%$), Gaussian blur ($k = 3$), and Gaussian noise (variance 10–50). Gaussian noise was selected as a conservative perturbation to emulate sensor-level variability. Other noise models (e.g., impulse or multiplicative noise) were not pursued to avoid introducing unrealistic artifacts that could distort fine-grained blade textures. After each perturbation, the background was reset to a pure white background to prevent color leakage. Random scaling and coarse dropout were deliberately excluded to maintain morphological fidelity.

To balance the representation across classes, each original image generated at least two augmented variants, with additional oversampling of minority categories until parity with the largest class was achieved. This procedure produced balanced datasets with realistic variations in blade appearance while maintaining domain consistency through constant white backgrounds. Representative examples are shown in Fig. 3.

2.5. Model architecture

The proposed architecture, KerisRDNet, extends InceptionResNetV2 pretrained on ImageNet by incorporating two IRD blocks. The convolutional layers of the backbone were frozen during the initial training phase, while the IRD blocks were designed to capture the elongated geometry and fine *pamor* textures of the keris blades by enlarging the receptive fields without additional pooling and improving the multiscale sensitivity.

Each IRD block applied four parallel depthwise-separable convolutions, 1×1 (dilation 1), 3×3 (dilation 2), 3×3 (dilation 3), and 5×5 (dilation 4), with pre-activation (BatchNorm, ReLU) in each branch. The outputs were concatenated, followed by dropout ($p = 0.2$), residual scaling ($\times 0.1$), and a skip connection with an optional 1×1 projection when the channels differed. A squeeze-and-excitation module (reduction ratio 16) recalibrated the channel weights, further enhancing the discriminative capacity.

The KerisRDNet classifier head comprised GlobalAveragePooling, Dense(512)–BatchNorm–ReLU–Dropout(0.5), Dense(256)–BatchNorm–ReLU–Dropout(0.5), and a task-specific softmax output. The internal design of an IRD block is shown in Fig. 4, and the stage-wise tensor map is presented in Table 3.

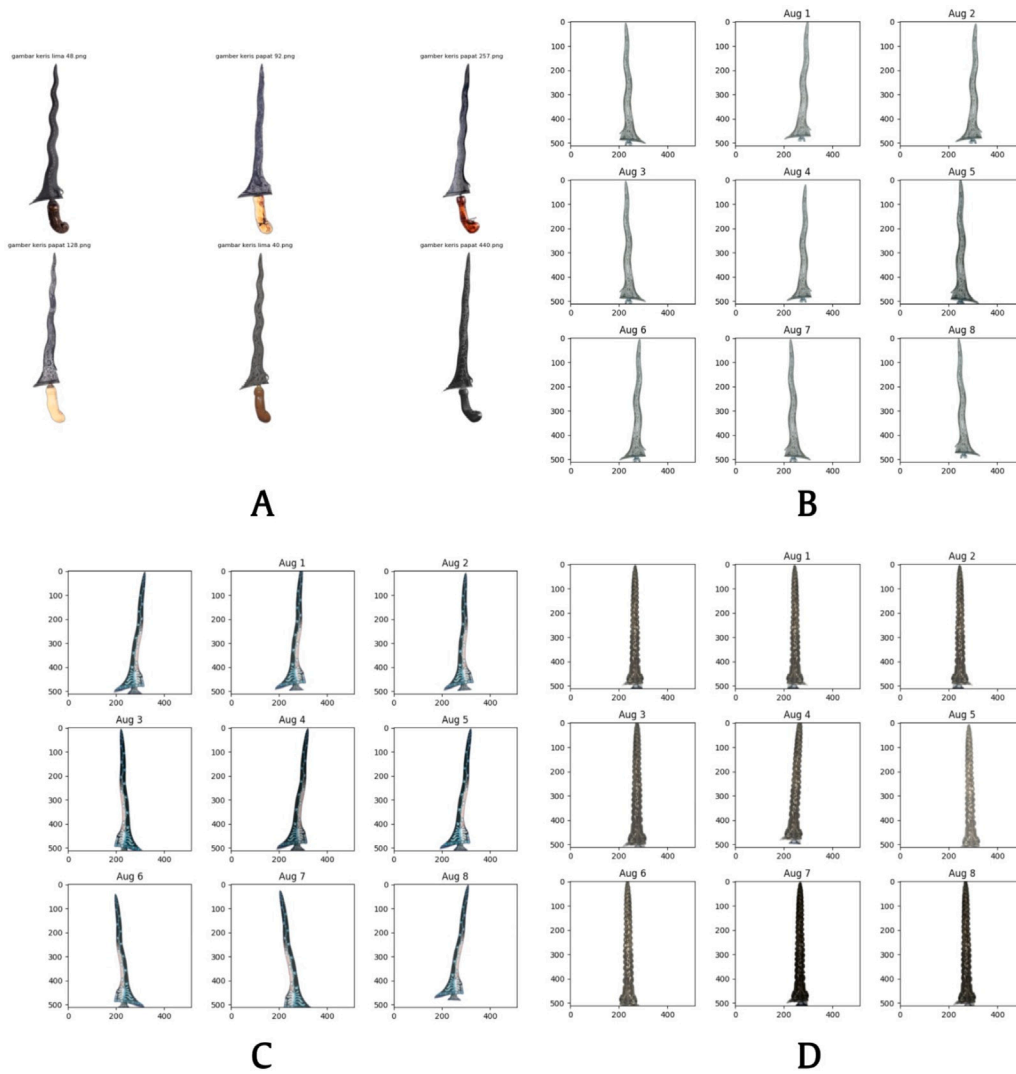


Fig. 3. Representative augmentation examples: (A) raw keris photographs; (B, C, D) mask-aware augmented outputs for *pamor*, *dhapur*, and *tangguh*. Geometric transforms were limited and photometric perturbations were restricted to blade regions, with backgrounds enforced as pure white.

Table 3

Stage-wise tensor map of KerisRDNet (input $512 \times 512 \times 3$). The architecture extends InceptionResNetV2 with two IRD blocks before the classifier head.

Stage	Operation	Tensor (H×W×C)
Input	–	$512 \times 512 \times 3$
Backbone output	InceptionResNetV2 (frozen, stride 32)	$\approx 16 \times 16 \times 1536$
IRD block #1 ^a	Inception–Residual–Dilated, filters = 768	$16 \times 16 \times 768$
IRD block #2	Same as IRD #1	$16 \times 16 \times 768$
Pooling	GlobalAveragePooling2D	768
FC1	Dense 512 + BN + Dropout 0.5 (ReLU)	512
FC2	Dense 256 + BN + Dropout 0.5 (ReLU)	256
Output	Dense n (Softmax)	$n \in \{27, 42, 5\}$

^a IRD block: four separable branches (dilations 1, 2, 3, 4), concatenation → dropout → residual scaling; skip connection with optional 1×1 projection; SE module (reduction 16).

precision and applied gradient clipping of 1.0. Models were optimized with Adam under an exponential learning-rate decay, initialized at 5×10^{-5} and decayed by 4% every 10^5 steps. Focal loss ($\alpha = 0.25$, $\gamma = 2.0$) with class weights proportional to the label frequencies was employed to address the class imbalance. Training ran for up to 100 epochs with a batch size of 8, early stopping (patience 8), and ReduceLROnPlateau (factor 0.5, patience 4) while monitoring validation macro-F1. The best model was checkpointed using the validation macro-F1. Random seeds were fixed across Python, NumPy, and the framework to ensure reproducibility. All experiments were repeated with three stratified resamplings of the train/validation split while keeping the held-out test set fixed, and results were reported on that fixed test set. Experiments were conducted on Ubuntu 22.04.3 LTS (kernel 5.4.0–155) with 2× AMD EPYC 7742 CPUs (256 threads) and 1.0 TiB RAM. Training used a single NVIDIA A100-SXM GPU (40 GB class; single-GPU allocation) with driver 525.147.05 and CUDA 12.2 (nvcc 12.2.140). The implementation used Python 3.10.12, TensorFlow 2.20.0 (tf-keras 2.20.1) for KerisRDNet, and PyTorch 2.0.1+cu117 for PyTorch-based baselines.

2.6. Training procedure

As part of the modeling stage, KerisRDNet was trained using the following optimization setup: We used TensorFlow/Keras with mixed

2.7. Evaluation protocol

The evaluation stage of the pipeline comprised three components: (i) protocol, which defines how the models were trained and validated;

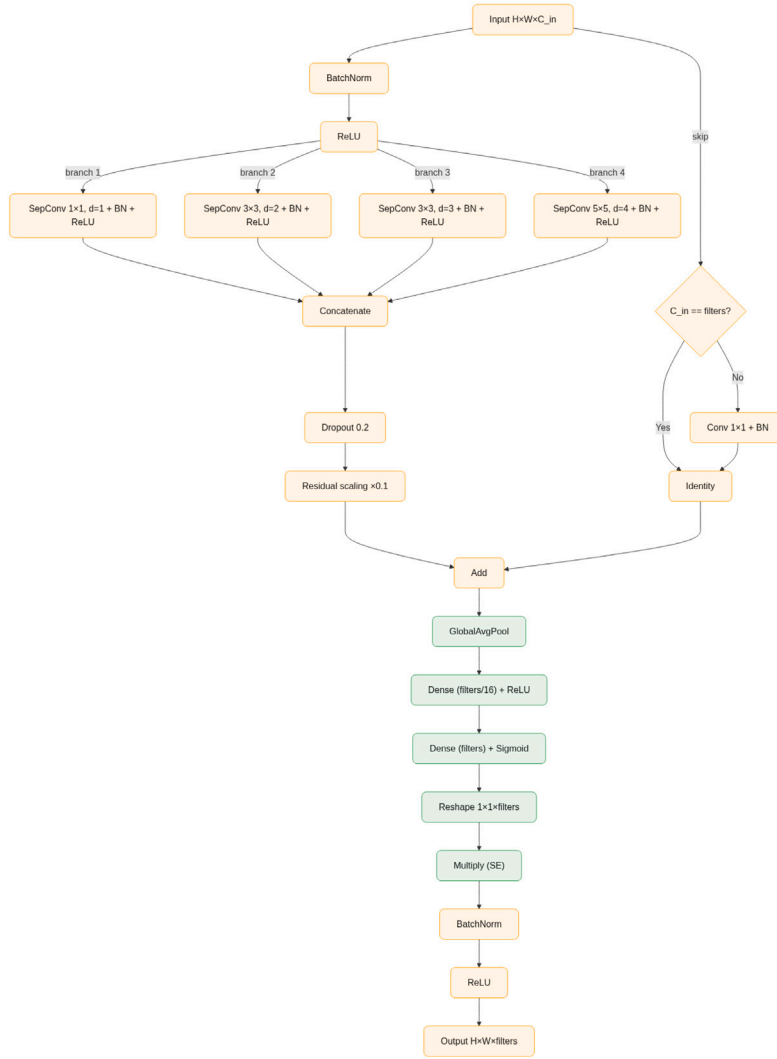


Fig. 4. Inception-Residual-Dilated (IRD) block used within KerisRDNet. Four depthwise-separable branches with dilations {1, 2, 3, 4} are concatenated, followed by dropout, residual scaling, and skip connection. A squeeze-and-excitation module (SE) recalibrates channels before BatchNorm and ReLU.

(ii) baselines and ablations, which establish controlled comparisons; and (iii) evaluation metrics, which quantify performance and statistical significance.

We trained one model per task (*pamor*, *dhapur*, *tangguh*) using the predefined train, validation, and test splits established at the artifact level (Table 1), ensuring that images of the same keris did not appear across partitions. The model selection was based on the highest validation macro-F1 within an early stopping window of eight epochs. All hyperparameters were fixed a priori, and no tuning was performed on the test sets. To assess split stability, training was repeated with three stratified resamplings of the train/validation split, and we report the mean $\pm$ standard deviation on the held-out test set.

2.8. Baselines and ablations

To isolate the contributions of architectural modifications and augmentation, we defined four controlled ablation settings based on InceptionResNetV2 as follows:

- (i) **Backbone-only:** InceptionResNetV2 with a classifier head, without IRD blocks or mask-aware augmentation.
- (ii) **IRD-only:** InceptionResNetV2 extended with IRD blocks, trained without augmentation.

(iii) **Aug-only:** Backbone-only model trained with mask-aware augmentation.

(iv) **KerisRDNet (proposed):** IRD-augmented backbone (IRD $\times$ 2) trained with mask-aware augmentation.

In addition to these ablations, we evaluated a set of strong contemporary baselines to ensure a fair comparison with modern architectures. These included a modern convolutional network (ConvNeXt-T), transformer-based models (ViT-B/16 and Swin-T), and large-scale pre-trained vision encoders (CLIP ViT-B/16) equipped with linear and shallow MLP classification heads. All baseline models were trained and evaluated under the same preprocessing pipeline (background homogenization, PCA-based orientation normalization, and YOLOv8 blade cropping), mask-aware augmentation strategy, artifact-level data splits, and training protocol as KerisRDNet, using pretrained weights and without model-specific hyperparameter tuning.

These controlled ablations and expanded baselines enable the assessment of (a) augmentation alone, (b) architectural adaptation alone, (c) their combined effect in KerisRDNet, and (d) the relative performance of KerisRDNet against strong modern CNN and transformer-based alternatives. Quantitative results and statistical validation are reported in Section 3.

Table 4

YOLOv8x (512 × 512) blade detector performance on the validation set.

Model	Precision	Recall	mAP ₅₀	mAP ₅₀₋₉₅
YOLOv8x	0.998	1.000	0.994	0.961

2.9. Evaluation metrics

The model performance was assessed using accuracy, precision, recall, F1-score, ROC–AUC, and Top-3 accuracy. Top-3 accuracy follows the standard top- k definition with $k = 3$. The ROC–AUC was computed in a one-vs-rest (OvR) manner and macro-averaged across classes. For statistical inference, we used the Wilcoxon signed-rank test as the primary non-parametric method, with paired t -tests reported for completeness. Given the small number of repetitions ($n = 3$), we report the Wilcoxon signed-rank test as the primary paired inference method, with paired t -tests included for completeness.

3. Results

3.1. Overview of experimental results

This section reports the empirical evaluation of the proposed pipeline, focusing on reliability, classification performance, and statistical consistency across resamplings. We organized the results into three stages that mirror the problem formulation in Section 1 and the experimental design in Section 2: (i) evaluation of blade region-of-interest (ROI) extraction as the basis of preprocessing, (ii) task-specific classification on *pamor*-27, *dhapur*-42, and *tangguh*-5 under controlled baselines and ablations, and (iii) cross-metric statistical validation and qualitative error analysis.

Consistent with the pronounced class imbalance and fine-grained nature of these tasks, macro-F1 is used as the primary endpoint, with accuracy, precision, recall, ROC–AUC, and Top-3 accuracy reported as complementary measures.

3.2. Performance of blade detection module

As motivated in Section 2, blade-focused cropping is required to suppress spurious background cues and focus on learning diagnostically relevant patterns. The YOLOv8x detector achieves near-perfect recall and very high precision on the validation split, providing reliable ROIs for downstream classification (Table 4). Representative qualitative examples are shown in Fig. 2.

As summarized in Table 4, the detector achieved near-perfect recall and very high precision with strong localization quality. These properties ensure reliable blade crops for downstream classification processes.

3.3. Results on *pamor* classification (27 classes)

Among the three tasks, *pamor* is the most challenging because of the subtle forging patterns and the large number of classes (27). As shown in Table 5, the baseline models collapsed with a macro-F1 score near zero. Mask-aware augmentation alone showed a modest but directionally consistent improvement across resamplings, corresponding to an average cross-metric gain of approximately +0.07 (Appendix Table C.1). IRD modification alone exhibited a larger paired effect ($\Delta \approx +0.11$, $d_z = 0.86$), though under $n = 3$ this is interpreted conservatively as directional consistency. The full pipeline (KerisRDNet) produced the largest and most consistent paired gains, raising the macro-F1 to 0.268 and AUC to 0.813 ($d_z = 3.24$, mean $\Delta \approx +0.27$). Top-3 accuracy of 0.570 further indicates decision-support potential, as the correct class often appears among the top three predictions. Fig. 5 visualizes these cross-metric differences, highlighting the collapse of backbones and the advantage of KerisRDNet, especially in macro-F1 and AUC.

Table 5*Pamor* (27 classes): test performance across baseline and proposed models.

Model	Loss	Acc	Prec	Rec	F1	AUC	Top-3
Backbone-only (InceptionResNetV2)	0.8257	0.0814	0.0000	0.0000	0.0000	0.5462	0.1163
IRD-only (InceptionResNetV2 + IRD)	0.7587	0.1628	0.3333	0.0116	0.0225	0.6569	0.2558
Aug-only (InceptionResNetV2 + Aug)	1.0674	0.1047	0.1579	0.0349	0.0571	0.6224	0.2791
KerisRDNet (IRD + Aug, proposed)	0.6216	0.3023	0.4146	0.1977	0.2677	0.8131	0.5698

Table 6*Dhapur* (42 classes): test performance across baseline and proposed models.

Model	Loss	Acc	Prec	Rec	F1	AUC	Top-3
Backbone-only (InceptionResNetV2)	0.9090	0.0120	0.0000	0.0000	0.0000	0.5310	0.0930
IRD-only (InceptionResNetV2 + IRD)	0.6160	0.2840	0.4860	0.1110	0.1810	0.8460	0.4640
Aug-only (InceptionResNetV2 + Aug)	1.1650	0.2220	0.2740	0.1910	0.2250	0.7670	0.4070
KerisRDNet (IRD + Aug, proposed)	0.6210	0.3210	0.4290	0.2040	0.2760	0.8530	0.5370

3.4. Results on *dhapur* classification (42 classes)

The *dhapur* task, which comprises 42 blade forms with subtle variations in curvature and guard features, is the most morphologically complex setting. As shown in Table 6, the baseline models collapsed, with a macro-F1 score near zero for the backbone-only variant. Augmentation alone yielded the strongest directionally consistent improvement across resamplings (see Appendix Table C.2), corresponding to a mean cross-metric gain of +0.228, demonstrating that data diversity is particularly beneficial in this case. IRD modification alone showed a large paired effect under the evaluated setting ($d_z = 1.91$); given $n = 3$, Wilcoxon signed-rank tests are interpreted conservatively as evidence of directional consistency rather than definitive inferential significance. The full pipeline (KerisRDNet) delivered the strongest and most consistent gains, achieving a macro-F1 of 0.276 and AUC of 0.853, with statistical validation ($p = 0.0011$, Cohen's $d_z = 3.77$) and an average improvement of +0.308 across all metrics. Top-3 accuracy of 0.537 further indicates that the model frequently narrows predictions to the correct family, even when the precise subclass is misidentified. As illustrated in Fig. 6, residual errors remain concentrated within sibling *dhapur* families, distinguished only by minor angular differences. These findings echo curatorial reports that *dhapur* recognition is difficult even for experts, underscoring the value of automated support tools and motivating richer geometric and multimodal representations in future work.

3.5. Results on *tangguh* classification (5 classes)

The *tangguh* task, which involves only five classes corresponding to broad historical eras, is less fine-grained than *pamor* or *dhapur*. As shown in Table 7, the baseline models achieved moderate performance, with macro-F1 values of approximately 0.38–0.42. Augmentation applied to the backbone did not yield consistent improvements across resamplings (see Appendix Table C.3), and an average cross-metric change of −0.003. IRD modification alone produced a larger paired gain ($\Delta = +0.046$, $d_z = 0.73$), which under $n = 3$ is interpreted conservatively as directional consistency rather than definitive significance. Only the full pipeline (KerisRDNet) achieved the most consistent gains, raising the macro-F1 to 0.635 and AUC to 0.805 ($d_z = 2.20$, mean $\Delta = +0.200$). The Top-3 accuracy of 0.824 was the highest across all tasks, indicating

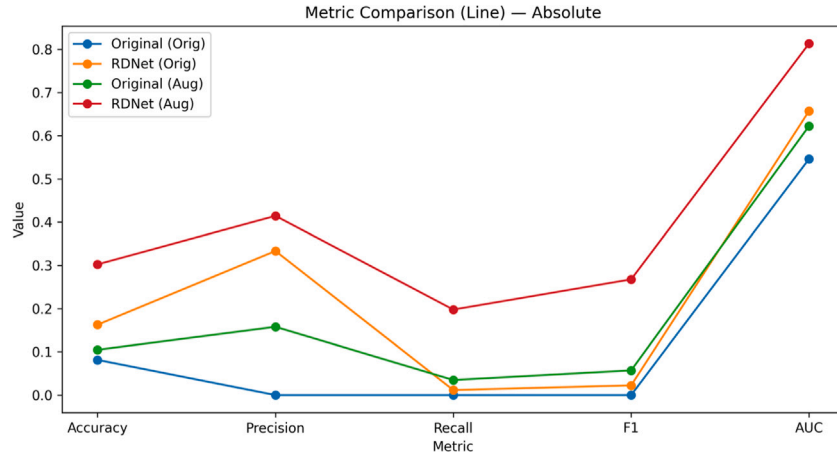


Fig. 5. *Pamor* (27 classes): cross-metric comparison of baseline and proposed models. KerisRDNet achieves the largest relative improvements, with clear gains in macro-F1 and AUC.

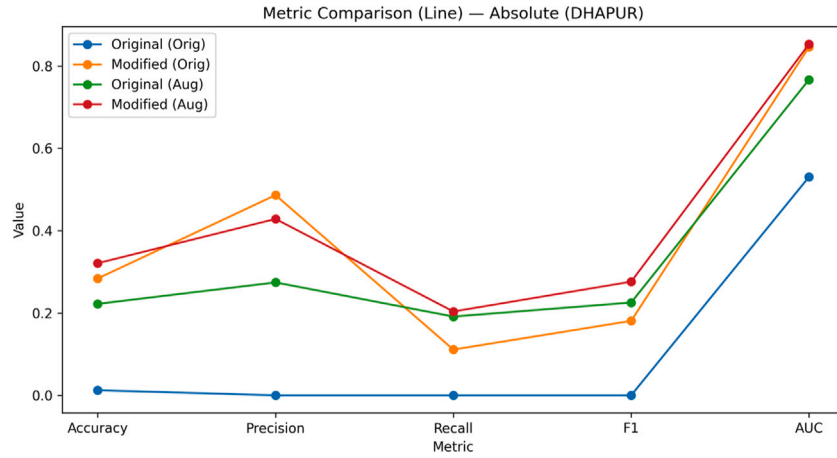


Fig. 6. *Dhapur* (42 classes): cross-metric comparison of baseline and proposed models. KerisRDNet delivers the strongest overall gains, with notable improvements in macro-F1 and AUC.

Table 7

Tanggung (5 classes): test performance across baseline and proposed models.

Model	Loss	Acc	Prec	Rec	F1	AUC	Top-3
Backbone-only (InceptionResNetV2)	0.2749	0.4412	0.4583	0.3235	0.3793	0.7631	0.8235
IRD-only (InceptionResNetV2 + IRD)	0.2159	0.4706	0.6111	0.3235	0.4231	0.7650	0.7941
Aug-only (InceptionResNetV2 + Aug)	0.4195	0.3824	0.4483	0.3824	0.4127	0.7238	0.7647
KerisRDNet (IRD + Aug, proposed)	0.3067	0.6471	0.6897	0.5882	0.6349	0.8055	0.8235

that the model reliably captured coarse stylistic distinctions relevant for provenance attribution. As visualized in Fig. 7, residual cross-era confusions were markedly fewer than in *pamor* and *dhapur*, reflecting that morphological and *pamor*-level cues provide sufficient discriminative power at this broader scale. These findings indicate that the proposed pipeline performs consistently across both fine- and coarse-grained tasks under the evaluated artifact-level setting, supporting its use as a decision-support tool in curatorial workflows.

3.6. Cross-metric statistical validation

To assess the stability of observed performance differences, we conducted paired comparisons across three stratified resamplings of the train/validation split while keeping the held-out test set fixed (Section 2). Given the small number of repetitions ($n = 3$), we report the Wilcoxon signed-rank test as the primary non-parametric paired test and include paired t -tests for completeness.

To avoid inflating Type-I error through separate hypothesis tests across multiple endpoints, statistical tests were applied to a cross-metric summary vector (Accuracy, Precision, Recall, macro-F1, and ROC-AUC), reported as averaged paired differences. Macro-F1 remains the primary performance indicator and is reported explicitly in the main results tables; however, under $n = 3$ the statistical results are interpreted conservatively as evidence of directional consistency rather than definitive effect magnitude. Accordingly, claims of improvement throughout Sections 3 and 4 are framed in terms of effect magnitude and directional consistency, rather than formal hypothesis rejection.

3.7. Qualitative error analysis

In addition to the aggregate metrics, we examined representative predictions to identify common success and failure modes. For the *pamor* task, KerisRDNet succeeded on high-contrast motifs such as *Beras Wutah* and *Udan Mas*; however, errors occurred when patterns were faint or partially corroded, leading to confusion with visually adjacent

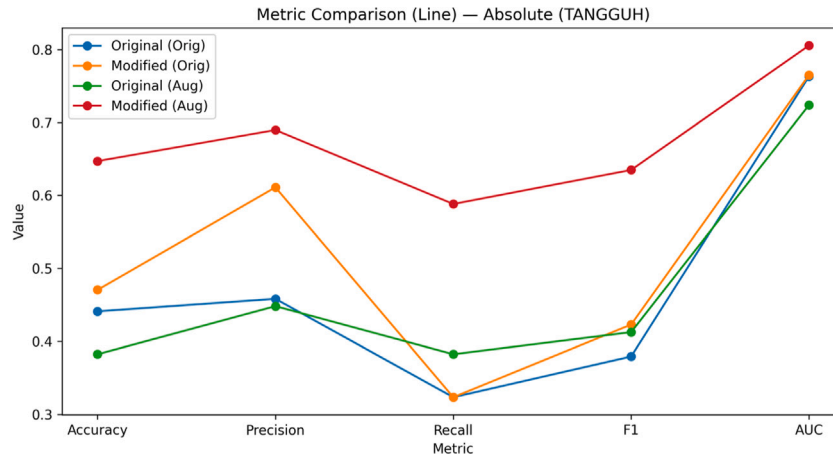


Fig. 7. *Tangguh* (5 classes): cross-metric comparison of baseline and proposed models. KerisRDNet shows the most consistent gains, particularly in macro-F1 and AUC.

classes. This aligns with the quantitative results, where macro-F1 remained low despite a relatively higher AUC, reflecting that visually similar motifs are difficult to separate, even for experts.

In the *dhapur* task, predictions were accurate for distinctive forms such as *Brojol* and *Tilam Upih*, while misclassifications were concentrated among sibling families with only subtle differences in curvature or guard design. These fine-grained confusions explain the low macro-F1, even though the Top-3 accuracy indicates that the correct family is often captured among the top predictions.

For the *tangguh* task, the model produced stable predictions for broad historical categories such as *Mataram*, with occasional misclassifications where modern reproductions closely mimic older styles (e.g., *Mataram* vs. *Kartasura*). Similar errors also appeared between *Mataram* and *Majapahit*. This pattern is consistent with the highest Top-3 accuracy among all tasks, confirming that even when the top label was incorrect, the correct class usually appeared among the top three possibilities.

Together, these qualitative patterns reinforce the quantitative findings: the proposed pipeline directs learning toward diagnostically salient blade features, mitigates but does not fully resolve, confusion among visually adjacent categories, and provides robust support under the evaluated experimental conditions.

3.8. Summary of results

Across all experiments, the proposed pipeline consistently outperformed the baseline configurations under both fine- and coarse-grained settings. The blade detection module achieved near-perfect recall and very high precision, ensuring reliable region-of-interest extraction for downstream classification. On the most challenging fine-grained tasks, *pamor* and *dhapur*, baseline models largely collapsed with macro-F1 scores near zero, whereas the integration of mask-aware augmentation and IRD blocks yielded substantial and directionally consistent gains, raising macro-F1 to 0.268 and 0.276, respectively, with improvements interpreted conservatively given the limited number of repetitions.

On the coarser *tangguh* task, where stylistic distinctions are broader, the proposed pipeline achieved the highest absolute performance, with a macro-F1 of 0.635 and Top-3 accuracy of 0.824, indicating robust discrimination across historical eras. Cross-metric statistical validation confirmed that these improvements were consistent across resamplings and not attributable to random variation, with directionally consistent paired differences across resamplings (Appendix Tables C.1–C.3), interpreted conservatively given $n = 3$. Qualitative error analysis further showed that residual misclassifications were concentrated among visually adjacent motifs and morphologically similar blade forms, reflecting

intrinsic ambiguities even for expert curators. These results demonstrate that the proposed pipeline provides a reliable and empirically validated foundation under conservative statistical interpretation for automated keris classification across varying levels of granularity.

3.9. Comparison with modern baselines

To complement the ablation analysis, we further compared KerisRDNet with strong modern baselines representing contemporary convolutional networks, transformer-based architectures, and large-scale pretrained vision encoders. This comparison evaluates whether the observed gains persist when contrasted against competitive alternatives.

As shown in Table 8, KerisRDNet achieves consistently strong macro-F1 across all three tasks, while several modern baselines achieve competitive ROC–AUC values that do not consistently translate into balanced class-wise performance.

Fig. 8 visualizes the relationship between ROC–AUC and macro-F1 aggregated across tasks. Models with high AUC but lower macro-F1 exhibit less balanced precision–recall behavior, whereas KerisRDNet maintains a more favorable balance between separability and class-wise discrimination. This visualization highlights the differing trade-offs between separability and balanced class-wise discrimination among the evaluated models.

4. Discussion

4.1. Key insights from the study

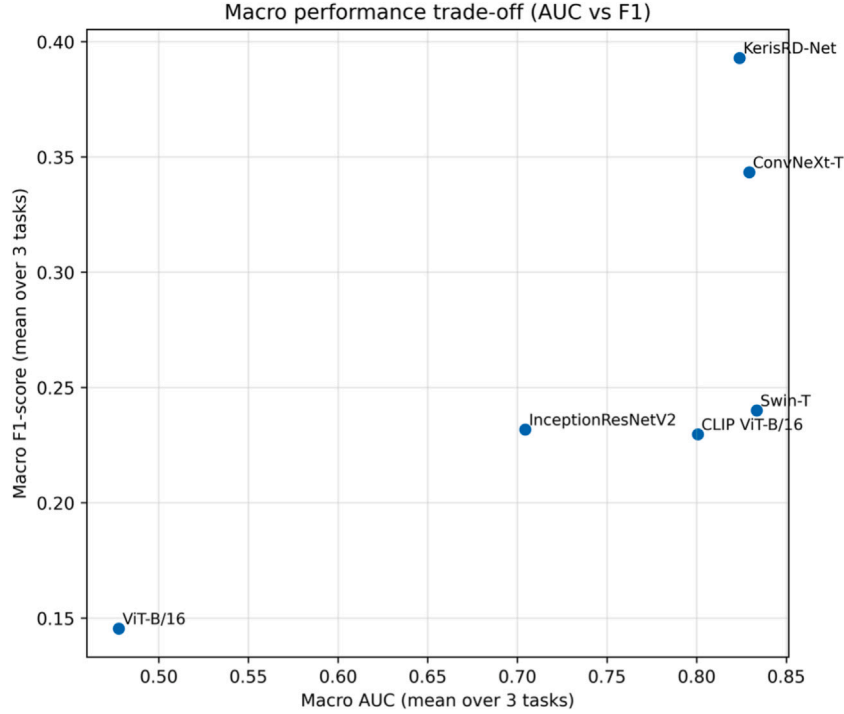
The findings of this study can be interpreted in direct relation to the four contributions outlined above. First, the proposed preprocessing workflow was essential for transforming heterogeneous keris imagery into a reproducible input domain. Background homogenization through `rembg` and white canvas composition eliminated contextual shortcuts, whereas PCA-based orientation with handle-down correction standardized the pose across artifacts. YOLOv8-based cropping further concentrated the learning on the blade, ensuring that the diagnostic regions, rather than the acquisition noise, formed the basis of the classification. These steps collectively stabilized the training and prevented the collapse of the standard backbones observed in the *pamor* and *dhapur* tasks.

Second, the mask-aware augmentation pipeline balanced the class distributions and enriched visual diversity while preserving morphological integrity. By confining geometric transformations (small shift-scale-rotate and light perspective) and photometric perturbations (contrast, blur, and noise) to the blade region, the method increased data variation without introducing spurious cues. This strategy is particularly

Table 8

Modern baseline comparison across tasks (macro-F1 and ROC-AUC), with macro-averages.

Model	Pamor		Dhapur		Tangguh		Macro-Avg	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC
KerisRDNet (proposed)	0.268	0.813	0.276	0.853	0.635	0.806	0.393	0.824
InceptionResNetV2 (Aug)	0.057	0.622	0.226	0.767	0.413	0.724	0.232	0.704
ViT-B/16 (Aug)	0.069	0.488	0.038	0.431	0.329	0.515	0.145	0.478
Swin-T (Aug)	0.263	0.830	0.153	0.854	0.304	0.816	0.240	0.833
ConvNeXt-T (Aug)	0.267	0.779	0.227	0.862	0.536	0.847	0.343	0.829
CLIP ViT-B/16 (Aug)	0.169	0.838	0.160	0.836	0.360	0.728	0.230	0.801

**Fig. 8.** Trade-off between ROC-AUC and macro-F1 for modern baselines and KerisRDNet (macro-averaged over tasks).

effective for fine-grained tasks, where augmentation substantially improves the macro-F1 score and reduces overfitting, confirming its role as a fairness mechanism under class imbalance (Chen et al., 2024).

Third, the proposed KerisRDNet architecture, which integrates IRD blocks with residual scaling and squeeze-and-excitation, provides a clear advantage over unmodified backbones. The design was well suited to the elongated geometry of the keris blades and subtle forging patterns of the *pamor* textures, capturing multiscale features that conventional convolutional networks failed to model effectively. These adaptations consistently delivered higher macro-F1, AUC, and Top-3 accuracy across tasks, demonstrating the value of architectural alignment with domain morphology.

Finally, a systematic evaluation of *pamor* (27 classes), *dhapur* (42 classes), and *tangguh* (5 classes) confirmed the robustness of our pipeline within the evaluated artifact-level setting. Across all three settings, the combined effect of preprocessing, augmentation, and KerisRDNet produced consistent improvements over strong baselines under the evaluated artifact-level setting, interpreted conservatively given the limited number of repetitions.

These results highlight the reproducibility and fairness of the approach, while also underscoring its practical readiness for deployment in digital heritage contexts. Taken together, this study positions keris recognition as a representative instance of fine-grained classification under low-data, high-variability conditions (Wei et al., 2022), while also offering a transferable framework for intelligent systems in other specialized domains.

4.2. Contribution of preprocessing

Preprocessing plays a decisive role in stabilizing performance and ensuring reproducibility under heterogeneous acquisition conditions. In this study, preprocessing comprises four conceptually distinct components: background homogenization, PCA-based orientation normalization, YOLOv8-based blade cropping, and mask-aware augmentation. While these steps are applied sequentially in the proposed pipeline, their individual contributions are not assumed to be uniform across tasks.

To explicitly quantify their effects, we conducted a controlled incremental ablation in which preprocessing components were introduced one at a time while keeping the backbone architecture and training protocol fixed. For each task, five configurations were evaluated, ranging from no preprocessing to the full pipeline with augmentation. The full per-configuration results (Models 1–5), including loss, accuracy, precision, recall, macro-F1, and AUC, are reported in Appendix B (Table B.1, Table B.2, and Table B.3).

From these primary ablation tables, we derive step-wise deltas in macro-F1 and AUC to provide a compact cross-task comparison of each preprocessing component. Importantly, these tables do not introduce additional experiments; they summarize step-wise differences computed from the per-configuration ablation results.

Table 9 summarizes the step-wise deltas in macro-F1, which reflect per-class decision quality under class imbalance, whereas Table 10 reports the corresponding deltas in AUC, capturing improvements in ranking separability.

Table 9

Incremental contribution of preprocessing components to macro-F1, derived from the ablation results in Appendix B.

Task	Background homogenization	Orientation normalization	Blade cropping	Mask-aware augmentation	Mean Δ
Tangguh	-0.023	+0.027	+0.215	+0.212	+0.108
Pamor	-0.006	+0.040	-0.019	+0.245	+0.065
Dhapur	+0.002	+0.088	+0.084	+0.095	+0.067

Table 10

Incremental contribution of preprocessing components to AUC, derived from the ablation results in Appendix B.

Task	Background homogenization	Orientation normalization	Blade cropping	Mask-aware augmentation	Mean Δ
Tangguh	-0.073	-0.039	+0.229	+0.040	+0.039
Pamor	+0.026	+0.026	+0.143	+0.156	+0.088
Dhapur	+0.051	+0.173	+0.068	+0.007	+0.075

Tangguh. The *tangguh* task exhibits the strongest dependency on blade-focused preprocessing. As shown in Tables 9 and 10, YOLOv8-based blade cropping yields the largest improvements in both macro-F1 and AUC, indicating that tangguh recognition relies primarily on diagnostically salient blade regions rather than contextual cues. Mask-aware augmentation provides a further substantial gain in macro-F1, improving performance stability under heterogeneous imaging conditions within the curated dataset.

Pamor. For *pamor*, PCA-based orientation normalization produces a consistent improvement in macro-F1, reflecting the sensitivity of fine-grained forging patterns to pose variability. Blade cropping markedly increases AUC, indicating improved class separability, but slightly reduces macro-F1. This divergence suggests that while ranking quality improves, top-1 decisions remain unstable under severe class imbalance. Once mask-aware augmentation is applied, both macro-F1 and AUC increase substantially, demonstrating that balanced, blade-restricted perturbations are necessary to translate improved separability into reliable per-class decisions.

Dhapur. In the *dhapur* task, orientation normalization emerges as the most influential preprocessing component, consistent with the shape-driven nature of dhapur classification. Blade cropping and augmentation provide complementary gains, confirming that pose normalization is a prerequisite for effective learning, while region-of-interest focusing and controlled diversity further enhance generalization.

Cross-task insight. Across all tasks, background homogenization alone shows inconsistent effects, whereas orientation normalization and blade cropping tend to more consistently reduce intra-class variance by standardizing pose and isolating diagnostically relevant regions. Mask-aware augmentation is the most consistently beneficial component for macro-F1 across tasks, particularly under strong class imbalance, underscoring its central role in stabilizing fine-grained recognition performance.

These findings align with prior work demonstrating that segmentation and alignment can improve reproducibility under uncontrolled imaging conditions (Singh et al., 2025; Wang et al., 2023). More broadly, they indicate that preprocessing is not a monolithic operation but a task-dependent intervention whose impact varies depending on whether classification is predominantly shape-driven, texture-driven, or provenance-oriented.

4.3. Effectiveness of augmentation and architectural adaptation

Mask-aware augmentation and architectural modification have emerged as complementary strategies for addressing class imbalances and fine-grained variability. On the data side, restricting transformations to blade regions while enforcing a constant white background prevented shortcut learning from the spurious cues. Small geometric perturbations (shift-scale-rotate and light perspective) and controlled

photometric variations (contrast, blur, and noise) enriched the representation of *pamor* textures without distorting their morphology. This design functions both as a generator of diversity and as a fairness mechanism for minority classes, improving generalization under low-data conditions (Johnson & Khoshgoftaar, 2019).

Quantitatively, augmentation delivered the strongest independent paired gains under the evaluated setting in the most fine-grained tasks. On *dhapur*, augmentation alone improved the macro-F1 by +0.227 with a large observed paired difference under the evaluated setting; given $n = 3$, we report paired statistics for completeness and interpret them conservatively. On *pamor*, the gains were smaller but directionally consistent (mean $\Delta \approx +0.07$; paired t -test $p \approx 0.043$), interpreted conservatively given $n = 3$. In contrast, the *tangguh* classification showed negligible benefit, reflecting its coarser granularity. This gradient confirms that augmentation is most valuable when the intra-class variation is subtle and the dataset imbalance is severe (Mumuni & Mumuni, 2022).

On the model side, the proposed KerisRDNet with IRD blocks provided further improvement. Its multi-branch dilated separable convolutions expanded the receptive fields without excessive pooling, residual scaling stabilized the training, and squeeze-and-excitation recalibration enhanced the channel sensitivity to subtle forging motifs. Modification alone yielded mean gains of +0.273 on *dhapur* ($d_z = 1.91$, $p = 0.013$) and +0.112 on *pamor* ($d_z = 0.86$), confirming the importance of architectural adaptation, even when preprocessing and augmentation were strong. These outcomes are consistent with reports in medical imaging, textile analysis, and fine-grained species recognition, where dilated convolutions and multiscale branches enhance discrimination under high intra-class similarity (Liu et al., 2022).

The expanded comparison with contemporary convolutional and transformer-based architectures provides additional context for these findings. While modern backbones and large-scale pretrained encoders can offer strong global representations and ranking capability, often reflected in competitive ROC-AUC values, such behavior does not necessarily translate into balanced class-wise discrimination when supervised adaptation data are limited. In our experiments, several modern baselines exhibited less stable precision-recall trade-offs, resulting in lower macro-F1 despite occasional gains in accuracy or AUC. In contrast, KerisRDNet integrates domain-aligned multiscale receptive fields, blade-focused preprocessing, and mask-aware augmentation, which together emphasize diagnostically salient regions and improve robustness across both minority and majority classes. This highlights that, under fine-grained and imbalanced cultural heritage settings, architectural and data-level alignment with object geometry can be as critical as overall model capacity.

Augmentation and architecture functioned synergistically; their combination in KerisRDNet produced consistent and statistically validated gains across macro-F1, AUC, and Top-3 accuracy. This demonstrates that domain-specific interventions at both the data and model levels are essential for fine-grained cultural heritage recognition.

4.4. Implications for digital heritage

In addition to improving accuracy and F1, the integrated pipeline has broader implications for digital heritage applications. First, the Top-3 accuracies above 0.5 across *pamor*, *dhapur*, and *tangguh* indicate that the system can function effectively as a decision-support tool. Even when top-1 predictions are uncertain, ranked outputs provide curators with plausible candidate sets for triage, cataloguing, and provenance assessment. This is especially valuable in heritage contexts, where expert resources are limited and inter-expert agreement is not always guaranteed (Foka & Griffin, 2024). Importantly, we position the proposed model as assistive rather than substitutive: it is intended to support expert workflows by narrowing plausible categories, while final interpretation and attribution remain with human experts.

Second, the reproducible preprocessing–augmentation–architecture workflow addresses reproducibility and fairness objectives, as emphasized in the Introduction. By enforcing standardized backgrounds, consistent orientation, balanced class distributions, and architecture tailored to the elongated blade geometry, the pipeline reduces noise and bias in model training. This structured approach offers a transferable template for other domains in which subtle textures and complex morphologies must be recognized under data scarcity, such as textiles, ceramics, and biomedical imaging. In this context, ethical and cultural governance is inseparable from technical reproducibility: keris images are culturally significant materials that remain under custodial authority, and any reuse or sharing must follow permission-based agreements consistent with curatorial and heritage protocols.

Third, the compact 512×512 design and lightweight classifier head make the system suitable for deployment on mobile and edge devices in the future. This design choice aligns with the focus of the resume pipeline on deployment readiness and resource efficiency. Comparable initiatives in intelligent systems suggest that such portability can democratize access to AI, enabling community-level engagement with cultural heritage while reducing dependence on centralized, high-performance infrastructure (Merizzi et al., 2024; Wang & Zhu, 2022). Together, these implications extend the contribution of KerisRDNet beyond academic benchmarks toward practical integration for digital heritage preservation and dissemination. Where expert disagreements are unavoidable in fine-grained attribution, our data curation process reflects this reality by reconciling labels through expert discussion and verification and excluding unresolved cases, thereby avoiding the propagation of uncertain ground truth into downstream models.

4.5. Generalizability and challenges

Although the proposed pipeline achieved consistent improvements across both fine- and coarse-grained tasks, several challenges remain. Confusion persisted in visually adjacent classes, reflecting the intrinsic difficulty of discriminating subtle morphological or textural differences, even for experts. Although the dataset was carefully curated and augmented, it remained modest in size, which limited the model’s exposure to the full diversity of keris artifacts. Robustness to domain shifts, such as variations in lighting, imaging hardware, or surface degradation, was not directly evaluated. Preprocessing and augmentation reduce but cannot fully eliminate these risks, indicating that broader digitization campaigns and cross-institutional validation are required to ensure wide applicability.

Despite these constraints, this study positions keris recognition as a proxy for fine-grained classification in low-data, high-variability regimes. This perspective highlights that the challenges observed here are not unique to cultural heritage but resonate with other intelligent system applications under similar conditions. Therefore, future work should pursue multi-task learning that jointly models *pamor*, *dhapur*, and *tangguh*, explore interpretable attention mechanisms to enhance transparency for curators, and investigate self-supervised or transfer learning to strengthen generalization under data scarcity. Federated

and cross-institutional approaches may also enable scaling while respecting custodial restrictions. These directions point toward more robust, interpretable, and ethically responsible intelligent systems for cultural heritage and related fields.

To support reproducibility, we provide a public repository containing the implementation and configuration files for the preprocessing, augmentation, training, and evaluation pipeline, including runnable scripts that export the main quantitative results (e.g., CSV metrics summaries and confusion matrices) for table reproduction. Due to cultural custodianship and governance constraints, the full dataset cannot be publicly released; the repository therefore documents the required local data layout and provides an optional small anonymized sample subset where permitted by the custodians. Where allowed, trained weights are also provided via the repository. Third-party tools and their licenses/terms (including rembg and YOLOv8/ultralytics) are listed in a dedicated third-party notices file in the repository. Repository: <https://github.com/masaboe/Keris-RDIN>.

5. Limitations and future work

This study also highlights several areas for improvement that frame directions for future research. Although mask-aware augmentation increased the dataset balance, the underlying corpus remained modest and could not capture the full diversity of keris artifacts or the range of degradations found in real-world collections. Residual ambiguities in visually adjacent classes, particularly among sibling *pamor* motifs and morphologically similar *dhapur* motifs, continue to pose challenges. In addition, the decision process of KerisRDNet remains difficult to interpret for non-technical users, and dataset sharing is constrained by cultural custodianship, which limits the opportunities for external benchmarking.

In addition, robustness under explicit curator- or acquisition-setting domain shifts was not evaluated in this study through held-out acquisition-source splits or controlled perturbation experiments (e.g., lighting or sensor changes). Curator-held-out evaluation and calibration assessment under such shifts (e.g., ECE/Brier) are therefore identified as important directions for future cross-institutional validation when acquisition-source annotations become available.

Future work can address these challenges in three main directions.

- **Methodological:** Multi-task learning frameworks that jointly model *pamor*, *dhapur*, and *tangguh* could leverage shared features across categories, while interpretable attention-based visualization would improve transparency for curators.
- **Architectural:** Benchmarking KerisRDNet against transformer-based models such as ViT or Swin may reveal complementary strengths in capturing long-range dependencies and subtle surface patterns.
- **Scalability and sustainability:** Efficiency analyses are needed to assess trade-offs between accuracy and computational cost, and federated or cross-institutional learning strategies offer a pathway to scaling without compromising data ownership or cultural sensitivities.

6. Conclusion

This study presents a domain-specific deep learning pipeline for keris classification as a case of fine-grained cultural heritage recognition. The pipeline combined standardized preprocessing, mask-aware augmentation, and the proposed KerisRDNet architecture tailored with IRD blocks and a squeeze-and-excitation mechanism. Together, these components stabilized the inputs, mitigated the imbalance, and improved the discriminative capacity where the generic backbones failed.

Across *pamor* (27 classes), *dhapur* (42 classes), and *tangguh* (5 classes), the pipeline achieved directionally consistent gains across resamplings, supported by paired analyses, interpreted conservatively

given $n = 3$, raising the macro-F1 well above near-zero baselines and delivering Top-3 accuracies above 0.5. These outcomes demonstrate the practical value of the proposed model as a decision-support tool for curators, reducing reliance on manual expertise while preserving human oversight.

Beyond keris, the workflow offers a transferable strategy for low-data fine-grained recognition tasks, balancing accuracy, reproducibility, and deployment efficiency under the evaluated experimental setting. By aligning domain-specific engineering with rigorous evaluation, KerisRDNet contributes both a functional tool for cultural preservation and a generalizable approach for fine-grained classification under data scarcity.

CRedit authorship contribution statement

Khafiizh Hastuti: Conceptualization, Supervision. **Erwin Yudi Hidayat:** Data curation, Writing – original draft. **Abu Salam:** Software, Writing – review & editing. **Usman Sudibyo:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Dataset class distributions

This appendix reports the per-class distributions for the three tasks. “Augmented Train” reflects the training size after mask-aware augmentation and class balancing. The validation and test sets were held out and unaugmented (see Table A.2).

Appendix B. Incremental ablation of preprocessing components

Table A.1

Per-class distribution for *pamor* classification (27 classes).

Class	Label	Train	Augmented train	Validation/Test
<i>Adeg</i>	0	92	276	14/14
<i>Beras Wutah</i>	1	15	117	2/2
<i>Blarak Sineret</i>	2	7	117	1/1
<i>Bonang Serenteng</i>	3	7	117	1/1
<i>Dwiwarna</i>	4	8	117	1/1
<i>Ganggeng Kanyut</i>	5	12	117	1/2
<i>Keleng</i>	6	6	117	1/1
<i>Kulit Semangka</i>	7	5	117	1/1
<i>Lar Gangsir</i>	8	7	117	1/1
<i>Melati Rinonce</i>	9	8	117	1/1
<i>Motif Ngulit Semangka</i>	10	14	117	2/2
<i>Nggajih</i>	11	8	117	1/1
<i>Nunggak Semi</i>	12	10	117	2/1

(continued on next page)

Table A.1 (continued).

Class	Label	Train	Augmented train	Validation/Test
<i>Pedaringan Kebak</i>	13	58	174	9/8
<i>Pengawak Wojo</i>	14	6	117	1/1
<i>Pulo Tirto</i>	15	9	117	1/2
<i>Ron Gendhuru</i>	16	8	117	1/1
<i>Sanak</i>	17	5	117	1/1
<i>Tirto Tumetes</i>	18	15	117	3/2
<i>Triman</i>	19	117	351	17/18
<i>Tunggak Semi</i>	20	50	150	7/8
<i>Udan Mas</i>	21	6	117	1/1
<i>Uler Lulut</i>	22	8	117	1/1
<i>Untu Walang</i>	23	6	117	1/1
<i>Wahyu Tumurun</i>	24	8	117	1/2
<i>Wengkon</i>	25	69	207	11/10
<i>Wos Wutah</i>	26	10	117	2/1

Table A.2

Per-class distribution for *dhapur* classification (42 classes).

Class	Label	Train	Augmented train	Validation/Test
<i>Bethok</i>	0	6	69	2/2
<i>Brojol</i>	1	69	207	23/23
<i>Carang Soka</i>	2	5	69	2/2
<i>Carita Keprabon</i>	3	4	69	2/1
<i>Carubuk</i>	4	7	96	3/2
<i>Damar Murup</i>	5	5	69	1/2
<i>Jalak</i>	6	4	69	2/1
<i>Jalak Dinding</i>	7	4	69	1/1
<i>Jalak Ngore</i>	8	16	69	5/5
<i>Jalak Nyucup Madu</i>	9	4	69	1/1
<i>Jalak Sangu Tumpeng</i>	10	10	69	3/4
<i>Jalak Tilam Sari</i>	11	21	69	7/7
<i>Jalak Tilam Upih</i>	12	4	69	2/1
<i>Jangkung</i>	13	7	69	2/3
<i>Karno Tinanding</i>	14	4	69	1/1
<i>Kebo Lajer</i>	15	10	69	3/3
<i>Kebo Teki</i>	16	4	69	1/2
<i>Megantoro</i>	17	4	69	2/1
<i>Naga</i>	18	4	69	1/2
<i>Naga Liman</i>	19	4	69	1/1
<i>Naga Siluman</i>	20	10	69	3/3
<i>Naga Temanten</i>	21	3	69	1/1
<i>Nogososo</i>	22	20	69	6/7
<i>Pandawa</i>	23	5	69	2/1
<i>Pandawa Cinarita</i>	24	7	69	2/3
<i>Pandawa Singo</i>	25	6	69	2/2
<i>Panji Anom</i>	26	4	69	1/1
<i>Parungsari</i>	27	20	69	6/7
<i>Pasopati</i>	28	13	69	4/5
<i>Pulang Geni</i>	29	5	69	2/1
<i>Sabuk Inten</i>	30	13	69	5/4
<i>Sempana</i>	31	23	69	8/8
<i>Sempaner</i>	32	11	69	4/4
<i>Sengkelat</i>	33	12	69	4/4
<i>Sepaner</i>	34	4	69	2/1
<i>Sepokal</i>	35	6	69	2/2
<i>Singo</i>	36	5	69	2/1
<i>Singo Barong</i>	37	32	69	11/11
<i>Sinom</i>	38	8	69	3/3
<i>Sinom Robyong</i>	39	4	69	1/2
<i>Tilam Sari</i>	40	11	69	4/4
<i>Tilam Upih</i>	41	67	201	22/22

Table A.3

Per-class distribution for *tangguh* classification (5 classes).

Class	Train	Augmented train	Validation	Test
<i>Era Baru</i>	47	198	6	6
<i>Kamardhikan</i>	56	132	7	7
<i>Madura</i>	54	162	7	7
<i>Majapahit</i>	44	168	5	6
<i>Mataram</i>	66	141	8	8

Table B.1*Pamor* (27 classes): preprocessing ablation (Models 1–5).

ID	Model configuration	Loss	Acc	Prec	Rec	F1	AUC
(5)	Modified InceptionResNetV2 + mask-aware augmentation	0.6216	0.3023	0.4146	0.1977	0.2677	0.8131
(4)	Background homogenization + orientation normalization + YOLO blade cropping	0.7587	0.1628	0.3333	0.0116	0.0225	0.6569
(3)	Background homogenization + orientation normalization (no YOLO cropping)	0.7902	0.0581	0.0661	0.0841	0.0418	0.5144
(2)	Background homogenization only	0.8082	0.0116	0.0008	0.0185	0.0016	0.4887
(1)	No preprocessing (baseline)	0.7836	0.0232	0.0247	0.0041	0.0071	0.4628

Table B.2*Dhapur* (42 classes): preprocessing ablation (Models 1–5).

ID	Model configuration	Loss	Acc	Prec	Rec	F1	AUC
(5)	Modified InceptionResNetV2 + mask-aware augmentation	0.620513	0.320988	0.428571	0.203704	0.276151	0.853249
(4)	Background homogenization + orientation normalization + YOLO blade cropping	0.616361	0.283951	0.486486	0.111111	0.180905	0.846352
(3)	Background homogenization + orientation normalization (no YOLO cropping)	0.768100	0.191400	0.375000	0.055600	0.096700	0.778300
(2)	Background homogenization only	0.879900	0.043200	0.006900	0.030300	0.009100	0.605100
(1)	No preprocessing (baseline)	0.892800	0.037000	0.004100	0.030600	0.007200	0.554100

Table B.3*Tangguh* (5 classes): preprocessing ablation (Models 1–5).

ID	Model configuration	Loss	Acc	Prec	Rec	F1	AUC
(5)	Modified InceptionResNetV2 + mask-aware augmentation	0.3067	0.6471	0.6897	0.5882	0.6349	0.8055
(4)	Background homogenization + orientation normalization + YOLO blade cropping	0.2159	0.4706	0.6111	0.3235	0.4231	0.7650
(3)	Background homogenization + orientation normalization (no YOLO cropping)	0.2581	0.2941	0.1826	0.2726	0.2082	0.5360
(2)	Background homogenization only	0.2582	0.2353	0.1500	0.2357	0.1817	0.5748
(1)	No preprocessing (baseline)	0.3163	0.2353	0.1950	0.2250	0.2049	0.6481

Table C.1*Pamor* (27 classes): paired cross-metric statistical validation (Accuracy, Precision, Recall, macro-F1, ROC-AUC).

Comparison	Metrics	t-stat	p (t)	p (Wilcoxon)	Cohen's d_z	Mean Δ
Aug-only vs Backbone	5-metric avg	-2.934	0.043	0.063	1.31	+0.070
IRD-only vs Backbone	5-metric avg	-1.919	0.127	0.063	0.86	+0.112
KerisRDNet (IRD+Aug) vs Backbone	5-metric avg	-7.247	0.0019	0.063	3.24	+0.274

Table C.2*Dhapur* (42 classes): paired cross-metric statistical validation (Accuracy, Precision, Recall, macro-F1, ROC-AUC).

Comparison	Metrics	t-stat	p (t)	p (Wilcoxon)	Cohen's d_z	Mean Δ
Aug-only vs Backbone	5-metric avg	-16.302	0.000083	0.063	7.29	+0.228
IRD-only vs Backbone	5-metric avg	-4.264	0.0130	0.063	1.91	+0.273
KerisRDNet (IRD+Aug) vs Backbone	5-metric avg	-8.437	0.0011	0.063	3.77	+0.308

Table C.3*Tangguh* (5 classes): paired cross-metric statistical validation (Accuracy, Precision, Recall, macro-F1, ROC-AUC).

Comparison	Metrics	t-stat	p (t)	p (Wilcoxon)	Cohen's d_z	Mean Δ
Aug-only vs Backbone	5-metric avg	0.145	0.892	1.000	≈ 0	-0.003
IRD-only vs Backbone	5-metric avg	-1.625	0.180	0.068	0.73	+0.046
KerisRDNet (IRD+Aug) vs Backbone	5-metric avg	-4.913	0.0080	0.063	2.20	+0.200

Appendix C. Supplementary material

This Supplementary Material reports detailed paired cross-metric statistical analyses across three stratified resamplings ($n = 3$). Given the small number of repetitions, Wilcoxon signed-rank tests are interpreted conservatively as directional consistency rather than definitive effect magnitude.

Data availability

Data will be made available on request.

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